

CCNA 200-301 - Chapter 18: RIP Protocol Notes

Dynamic Routing

- Interior: RIP, OSPF, EIGRP
- Exterior: BGP
- RIP uses Hop Count
- OSPF uses Cost/Bandwidth
- EIGRP uses Bandwidth + Delay + Reliability

RIP Versions

- RIPv1: Classful, Broadcast, no VLSM
- RIPv2: Classless, VLSM, Multicast 224.0.0.9
- RIPng: IPv6

Updates

Periodic Update: Every 30 seconds

Triggered Update: Immediately after topology changes

Commands

```
router rip
```

```
version 2
```

```
network 192.168.1.0
```

```
network 11.0.0.0
```

```
network 12.0.0.0
```

```
show ip route
```

```
show ip protocols
```

```
wr
```

show ip route

```
R 200.200.200.0/24 [120/2] via 10.0.0.2
```

- 120 = Administrative Distance
- 2 = Hop Count
- via = Next Hop
- Max Hop Count = 15
- 16 Hop = Unreachable

- Routers advertise only directly connected networks